# Analytic Supplement for A Dirichlet Pólya Inequality on Three-Dimensional Spherical Shells

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17 July 2026

**Abstract**

This support volume contains the detailed global defect proof used for $0 < \rho \leq 39/50$. It proves the ratio-sharp angular estimate, the tangent-envelope retained-radial estimate, reduction to a universal ball quarter-layer, and the exact scalar convexity closure. Together with the phase bridge, no-mode comparison, and optical theorem in the main paper, this is the complete analytic implication chain for the low- and middle-ratio region. Every estimate used below is proved analytically; no program, floating-point computation, or interval certificate is a theorem premise.

**Reading convention.** The detailed argument is read directly from `analytic/ratio-sharp-global-closure-body.tex`; no precompiled PDF is required. A summarized form is included in the main paper. Every displayed rational bound below is part of the proof; numerical replay is not.

## Live dependency graph

| result | premises | location |
|---|---|---|
| phase-to-proxy bound | separation and strict Bessel-phase enclosure | main paper |
| no-mode region | one-dimensional Dirichlet comparison | main paper |
| global low/middle theorem | exact layer cake, ratio-sharp blocks, tangent envelope, ball quarter-layer, scalar convexity | Part I below |
| outer optical theorem | product and complementary-action screens | main paper |
| shell theorem | the preceding disjoint owners and dilation | main paper |

## Contents

# I Ratio-sharp global defect closure

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}.$$

Fix $0 < \rho < 1$ and $K > 0$. For $a > 0$, define the zero-extended ball action

$$G_a(z) := \begin{cases} \frac{1}{\pi}\left(\sqrt{a^2 - z^2} - z\arccos\frac{z}{a}\right), & 0 \le z < a, \\ 0, & z \ge a, \end{cases} \tag{1}$$

and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

Write

$$[u]_< := \#\{n \in \mathbb{N} : n < u\}.$$

The phase reduction in the main paper supplies

$$N_D(A_{\rho,1}, K^2) \le P(\rho, K), \qquad P(\rho, K) := \sum_{\substack{\nu=\ell+1/2<K \\ \ell\in\mathbb{N}_0}} 2\nu\,[A(\nu) + \tfrac{1}{4}]_<. \tag{3}$$

Indeed, the channel phase is strictly smaller than $A(\nu) + 1/4$, so at an integral proxy wall the last radial integer remains excluded. The continuous payment is

$$W(\rho, K) := \int_0^K 2zA(z)\,dz = \frac{2(1-\rho^3)K^3}{9\pi}. \tag{4}$$

**Action geometry and the exact defect**

Differentiating (1) gives

$$q(z) := -A'(z) = \frac{1}{\pi}\begin{cases} \arccos\frac{z}{K} - \arccos\frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos\frac{z}{K}, & \rho K \le z < K. \end{cases} \tag{5}$$

Thus $A$ decreases strictly from $T$ to 0. Let $R : [0, T] \to [0, K]$ be its decreasing inverse and set

$$F(t) := R(t)^2, \qquad g(t) := -F'(t), \qquad \tau := A(\rho K) = KG_1(\rho). \tag{6}$$

Write throughout

$$\theta := \arccos\rho.$$

**Lemma 1** (Shape and endpoint data)**.** *With this notation, one has*

$$0 \le q(z) \le \frac{\theta}{\pi}, \qquad q \text{ increases on } (0, \rho K), \qquad q \text{ decreases on } (\rho K, K), \tag{7}$$

*and*

$$g \text{ decreases on } (0, \tau), \qquad g \text{ increases on } (\tau, T). \tag{8}$$

*Moreover,*

$$F(\tau) = \rho^2K^2, \qquad g(\tau) = \frac{2\pi\rho K}{\theta}, \qquad g(T) = \frac{2\pi\rho K}{1-\rho}. \tag{9}$$

*The two formulas for $g$ agree at $t = \tau$, and $g(t) = O(t^{-1/3})$ as $t \downarrow 0$.*

*Proof.* Direct differentiation gives

$$q'(z) = \frac{1}{\pi}\left(\frac{1}{\sqrt{\rho^2K^2 - z^2}} - \frac{1}{\sqrt{K^2 - z^2}}\right) > 0 \quad (0 < z < \rho K),$$

whereas

$$q'(z) = -\frac{1}{\pi\sqrt{K^2 - z^2}} < 0 \quad (\rho K < z < K).$$

Both one-sided values at $z = \rho K$ equal $\theta/\pi$, proving (7).

If $z = R(t) > \rho K$, then

$$g(t) = \frac{2\pi z}{\arccos(z/K)}.$$

Its monotonicity follows from the explicit derivative

$$\frac{d}{dz}\left(\frac{z}{\arccos(z/K)}\right) = \frac{\arccos(z/K) + z/\sqrt{K^2 - z^2}}{\arccos(z/K)^2} > 0.$$

Since $z$ decreases with $t$, $g$ decreases on $(0, \tau)$. On the inner branch put

$$\delta(z) := \arccos\frac{z}{K} - \arccos\frac{z}{\rho K}.$$

The identity

$$\frac{\delta(z)}{z} = \int_\rho^1 \frac{ds}{s\sqrt{s^2K^2 - z^2}}$$

has derivative

$$\frac{d}{dz}\left(\frac{\delta(z)}{z}\right) = \int_\rho^1 \frac{z\,ds}{s(s^2K^2 - z^2)^{3/2}} > 0.$$

Hence $\delta(z)/z$ increases with $z$, and $g(t) = 2\pi z/\delta(z)$ increases as $t$ increases and $z$ decreases. The endpoint values follow by substitution and the limit $z \downarrow 0$. Finally, writing $z = K\cos u$ on the outer branch gives $t \asymp u^3$ and $g \asymp u^{-1}$. $\square$

Put

$$x_n := n - \frac{1}{4}, \qquad M(r) := \#\{\ell \in \mathbb{N}_0 : \ell + \tfrac{1}{2} < r\}, \qquad N := \#\{n \geq 1 : x_n < T\}. \tag{10}$$

For an active $n$, write $r_n := R(x_n)$ and $m_n := M(r_n)$.

**Lemma 2** (Strict layer cake)**.** *For every* $0 < \rho < 1$ *and* $K > 0$,

$$P(\rho, K) = \sum_{x_n < T} m_n^2, \qquad W(\rho, K) = \int_0^T F(t)\,dt. \tag{11}$$

*Both identities retain every radial and angular equality wall.*

*Proof.* For a half-integer $\nu = \ell + 1/2$,

$$[A(\nu) + \tfrac{1}{4}]_< = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since $A$ and $R$ are decreasing, this is equivalent to $\nu < R(x_n)$. Interchanging the finite sums and using $\sum_{\ell=0}^{m-1}(2\ell + 1) = m^2$ proves the first identity. At $R(x_n) = m + 1/2$, strict angular counting gives $M(R(x_n)) = m$. The second identity is the area formula $\int_0^T R(t)^2 dt = \int_0^K 2zA(z)dz$. $\square$

Define

$$\mathcal{E}_{\mathrm{ang}} := \sum_{x_n < T}(m_n^2 - r_n^2), \qquad \mathcal{D}_{\mathrm{rad}} := \int_0^T F(t)\,dt - \sum_{x_n < T} F(x_n). \tag{12}$$

Then

$$W - P = \mathcal{D}_{\mathrm{rad}} - \mathcal{E}_{\mathrm{ang}}. \tag{13}$$

**The ratio-sharp angular payment**

**Proposition 3** (Ratio-sharp angular blocks)**.** *For* $N \ge 1$*,*

$$\mathcal{E}_{\mathrm{ang}} < \frac{(1-\rho^2)K^2}{6} - N d(\rho), \qquad d(\rho) := \frac{3\pi}{8\arccos\rho} - \frac{1}{4}. \tag{14}$$

*The inequality is strict at every common radial–angular wall. If* $N = 0$*, then* $\mathcal{E}_{\mathrm{ang}} = 0$*.*

*Proof.* Fix an active $n$ and $t \in [n-1, x_n]$. Since $R(t) \ge r_n$,

$$x_n - t = A(r_n) - A(R(t)) = \int_{r_n}^{R(t)} q(s)\, ds \le \frac{\theta}{\pi}(R(t) - r_n).$$

Integration over the left block of length $3/4$ yields

$$r_n \le \frac{4}{3}\int_{n-1}^{x_n} R(t)\, dt - \frac{3\pi}{8\theta}. \tag{15}$$

Strict angular counting gives $m_n < r_n + 1/2$, including when $r_n$ is a half-integer, and hence

$$m_n^2 - r_n^2 < r_n + \frac{1}{4} \le \frac{4}{3}\int_{n-1}^{x_n} R(t)\, dt - d(\rho). \tag{16}$$

The blocks are disjoint subsets of $[0, T]$. Therefore

$$\mathcal{E}_{\mathrm{ang}} < \frac{4}{3}\int_0^T R(t)\, dt - N d(\rho).$$

The inverse area identity and scaling give

$$\int_0^T R(t)\, dt = \int_0^K A(z)\, dz = \frac{(1-\rho^2)K^2}{8},$$

which proves (14). □

**The tangent-envelope radial payment**

Let

$$C_0(t) := \#\{n \ge 1 : x_n < t\}, \quad w(t) := t - C_0(t), \quad \mathfrak{W}(t) := \int_0^t w(s)\, ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \tag{17}$$

On $0 \le t \le 3/4$,

$$\mathfrak{W}(t) = \frac{t^2}{2}, \qquad \Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \tag{18}$$

On every later unit cell, direct integration gives

$$-\frac{1}{32} \le \Psi(t) \le \frac{3}{32}, \qquad \mathfrak{W}(t) \ge \frac{t}{4} - \frac{1}{32}. \tag{19}$$

**Lemma 4** (Retained radial remainder)**.** *For all* $0 < \rho < 1$ *and* $K > 0$*,*

$$\mathcal{D}_{\mathrm{rad}} \ge \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) - \int_{(0,\tau)} \mathfrak{W}(t)\, dg(t). \tag{20}$$

*Proof.* Distributionally, $dw = dt - dC_0$. Stieltjes integration by parts gives $\mathcal{D}_{\mathrm{rad}} = \int_0^T w(t)g(t)dt$; the improper boundary term vanishes because $\mathfrak{W}(t)g(t) = O(t^{5/3})$ at zero. On the decreasing branch,

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_{(0,\tau)} \mathfrak{W}\, dg.$$

On the increasing branch use $w = 1/4 + \Psi'$, $\int_\tau^T g = F(\tau)$, and integrate by parts. Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, the interface value cancels and

$$\mathcal{D}_{\mathrm{rad}} = \frac{F(\tau)}{4} + \frac{\tau g(\tau)}{4} + \Psi(T)g(T) - \int_{(\tau,T)} \Psi\, dg - \int_{(0,\tau)} \mathfrak{W}\, dg.$$

Because $dg$ is positive on $(\tau, T)$,

$$\Psi(T)g(T) - \int_{(\tau,T)} \Psi\, dg \geq -\frac{g(T)}{8} + \frac{3g(\tau)}{32}.$$

This proves (20). Continuity of $g$ and $\mathfrak{W}$ handles an interface on a sawtooth wall without an atom. □

Define the $C^1$ tangent envelope

$$L_\sharp(t) := \begin{cases} t^2/2, & 0 \leq t \leq 1/4, \\ t/4 - 1/32, & t \geq 1/4, \end{cases} \qquad L_\sharp'(t) = \min\{t, 1/4\}. \tag{21}$$

Equations (18)–(19) show

$$0 \leq L_\sharp(t) \leq \mathfrak{W}(t) \qquad (t \geq 0). \tag{22}$$

On $1/4 \leq t \leq 3/4$ the difference is $\frac{1}{2}(t - 1/4)^2$; after 3/4 use (19).

Assume now

$$0 < \rho \leq \frac{39}{50}, \qquad K \geq \frac{\pi}{1-\rho}. \tag{23}$$

The elementary turning estimate

$$G_1(s) = \frac{1}{\pi}\int_s^1 \arccos u\, du \geq \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2} \tag{24}$$

is immediate: if $v = \arccos u$, then $1 - u = 1 - \cos v \leq v^2/2$, so $\arccos u \geq \sqrt{2(1-u)}$; integration over $[s, 1]$ gives the displayed constant. It implies

$$\tau \geq \frac{2\sqrt{2}}{3}\sqrt{1-\rho} \geq \frac{2\sqrt{2}}{3}\sqrt{\frac{11}{50}} > \frac{1}{4}.$$

The last inequality follows after squaring from $44/225 - 1/16 = 479/3600 > 0$.

On $(0, \tau)$, the measure $-dg$ is positive. Hence

$$\begin{aligned} -\int_{(0,\tau)} \mathfrak{W}\, dg &\geq -\int_{(0,\tau)} L_\sharp\, dg \\ &= -L_\sharp(\tau)g(\tau) + \int_0^\tau L_\sharp'(t)g(t)\, dt. \end{aligned} \tag{25}$$

There is no boundary contribution at zero. On the outer branch $t = A(z) = G_K(z)$ and $g(t)dt = -2z\, dz$, so

$$\int_0^\tau L_\sharp'(t)g(t)\, dt = \int_{\rho K}^K 2z \min\{G_K(z), 1/4\}\, dz. \tag{26}$$

Since $\tau > 1/4$, the interface factor becomes

$$g(\tau)\left(\frac{\tau}{4}+\frac{3}{32}-L_\sharp(\tau)\right)=\frac{g(\tau)}{8}.$$

Lemma 4 and (9) now give

$$\mathcal{D}_{\mathrm{rad}} \geq \mathcal{H}_\sharp(\rho, K), \tag{27}$$

where

$$\mathcal{H}_\sharp(\rho, K) := \frac{\rho^2K^2}{4} - \frac{\pi\rho K}{4(1-\rho)} + \frac{\pi\rho K}{4\arccos\rho} + \int_{\rho K}^{K} 2z\min\{G_K(z), 1/4\}\,dz. \tag{28}$$

The simplified interface payment is asserted only on (23). Below $\tau = 1/4$, its unsimplified factor is $g(\tau)(\tau/4 + 3/32 - \tau^2/2)$.

### The universal ball quarter-layer

Let $R_B$ be the decreasing inverse of the restriction of $G_K$ to $[0, K]$. Since $G_K(\rho K) = \tau > 1/4$, truncated layer cake gives

$$\frac{\rho^2K^2}{4} + \int_{\rho K}^{K} 2z\min\{G_K(z), 1/4\}\,dz = B_0(K) := \int_0^{1/4} R_B(t)^2\,dt. \tag{29}$$

Therefore

$$\mathcal{H}_\sharp(\rho, K) = B_0(K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{\pi\rho K}{4\arccos\rho}. \tag{30}$$

From (24) and $G_K(z) = KG_1(z/K)$,

$$R_B(t) \geq K - CK^{1/3}t^{2/3}, \qquad C := \left(\frac{3\pi}{2\sqrt{2}}\right)^{2/3} = \frac{(3\pi)^{2/3}}{2}. \tag{31}$$

The right side is nonnegative for $0 \leq t \leq 1/4$: it is enough that $K \geq C^{3/2}/4 = 3\pi/(8\sqrt{2})$, whereas $K \geq \pi/(1-\rho) > \pi$. Squaring and integrating gives

$$B_0(K) \geq \frac{K^2}{4} - \alpha K^{4/3} + \beta K^{2/3}, \tag{32}$$

where

$$\alpha := \frac{6C}{5\,4^{5/3}}, \qquad \beta := \frac{3C^2}{7\,4^{7/3}}. \tag{33}$$

Combining (13), (14), (27), and (32), and using $T \geq 1$, hence $N \geq 1$, gives

$$\begin{aligned} W - P > \frac{1+2\rho^2}{12}K^2 - \alpha K^{4/3} + \beta K^{2/3} - \frac{\pi\rho K}{4(1-\rho)} \\ + \frac{\pi\rho K}{4\arccos\rho} + d(\rho). \end{aligned} \tag{34}$$

### Exact scalar closure

**Lemma 5** (Elementary angle bound)**.** *For* $0 \leq \rho < 1$*,*

$$\arccos\rho \leq \frac{\pi}{2}\sqrt{1-\rho}. \tag{35}$$

*Proof.* Write $\rho = \cos\theta$, $0 \le \theta \le \pi/2$, and $y = \theta/2$. Since $\sin y/y$ decreases on $[0, \pi/4]$,

$$\frac{\sin y}{y} \ge \frac{2\sqrt{2}}{\pi}.$$

Indeed, $(\sin y/y)' = (y\cos y - \sin y)/y^2 < 0$, because $(\sin y - y\cos y)' = y \sin y > 0$. Using $1 - \rho = 2\sin^2 y$ gives (35). □

**Lemma 6** (Rational bounds for $\pi$)**.** *One has*

$$\frac{157}{50} < \frac{333}{106} < \pi < \frac{22}{7}.$$

*Proof.* Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. The tangent addition formula gives

$$\tan(2a) = \frac{5}{12}, \qquad \tan(4a) = \frac{120}{119}, \qquad \tan(4a - b) = 1.$$

Here $b < a$, while $(6/5)^2 < 2$ gives $1/5 < \sqrt{2} - 1 = \tan(\pi/8)$, and hence $a < \pi/8$. Therefore $0 < 4a - b < \pi/2$. Thus this is Machin's identity $\pi = 16a - 4b$. The alternating arctangent expansion gives

$$a > \frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5} - \frac{1}{7\,5^7}, \qquad b < \frac{1}{239}.$$

Indeed, these inequalities follow by integrating

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \frac{t^8}{1+t^2} \qquad (0 \le t \le 1)$$

and using $1/(1+t^2) < 1$ for $t > 0$. Thus

$$\pi > \frac{1231847548}{392109375} = \frac{333}{106} + \frac{3418213}{41563593750}, \qquad \frac{333}{106} - \frac{157}{50} = \frac{2}{1325} > 0.$$

For the upper bound,

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2},$$

so integration and $4\arctan 1 = \pi$ give

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

Also $22/7 - 333/106 = 1/742 > 0$, completing the asserted chain. □

Put $e := 1 - \rho$. It suffices to prove positivity of

$$\begin{aligned} \underline{\mathcal{G}}(\rho, K) := \frac{1+2\rho^2}{12}K^2 - \alpha K^{4/3} + \beta K^{2/3} - \frac{\pi\rho K}{4e} \\ + \frac{\rho K}{2\sqrt{e}} + \frac{3}{4\sqrt{e}} - \frac{1}{4}. \end{aligned} \tag{36}$$

Define

$$\mathsf{A} := \alpha\pi^{4/3}, \quad \mathsf{B} := \beta\pi^{2/3}, \quad C_1 := \frac{4}{3}\alpha\pi^{1/3}, \quad C_2 := \frac{2}{3}\beta\pi^{-1/3}. \tag{37}$$

Positive-side cubing and Lemma 6 give

$$\mathsf{A} < \frac{49}{40}, \qquad \frac{179}{1000} < \mathsf{B} < \frac{1}{5}, \qquad C_1 < \frac{13}{25}, \qquad C_2 < \frac{1}{25}. \tag{38}$$

For completeness,

$$\mathsf{A}^3 = \frac{3^5\pi^6}{5^3 2^{10}}, \quad \mathsf{B}^3 = \frac{3^7\pi^6}{7^3 2^{20}}, \quad C_1^3 = \frac{3^2\pi^3}{5^3 2^4}, \quad C_2^3 = \frac{3^4\pi^3}{7^3 2^{17}}. \tag{39}$$

Thus each comparison in (38) is an exact rational cross multiplication after cubing.

**Exact scalar reserve table.** For reproducibility, the positive margins used in those cubed comparisons and in the final convexity propagation are recorded below. The first five rows are respectively the rational difference between the cube of the stated upper bound and the certified upper cube, or between the certified lower cube and the cube of the stated lower bound.

| inequality | exact positive reserve |
|---|---|
| $\mathsf{A} < 49/40$ | 13125773/1505907200 |
| $\mathsf{B} > 179/1000$ | 521715693010963/5619712000000000000 |
| $\mathsf{B} < 1/5$ | 176853008713/82644187136000 |
| $C_1 < 13/25$ | 9767/10718750 |
| $C_2 < 1/25$ | 243014341/30118144000000 |
| turning-interface square above 1/16 | 479/3600 |
| $F_0'''$ lower bound | 676141/2450 |
| $D_0''$ lower bound | 1951/100 |
| $\partial_K^2 \underline{\mathcal{G}}$ lower bound | 47447/443682 |

Set

$$K_0 := \frac{\pi}{e}, \qquad x := e^{-1/6} \geq 1. \tag{40}$$

Direct substitution gives

$$\underline{\mathcal{G}}(\rho, K_0) = F_0(x), \tag{41}$$

where

$$\begin{aligned} F_0(x) := \frac{\pi}{2}x^9 - \mathsf{A}x^8 - \frac{\pi^2}{12}x^6 + \mathsf{B}x^4 \\ + \left(\frac{3}{4} - \frac{\pi}{2}\right)x^3 + \frac{\pi^2}{6} - \frac{1}{4}. \end{aligned} \tag{42}$$

The constant bounds imply

$$F_0(1) > \frac{8269}{30000} > \frac{1}{4}, \quad F_0'(1) > -\frac{14437}{6125} > -\frac{5}{2}, \quad F_0''(1) > \frac{18162}{1225} > 14. \tag{43}$$

For $x \geq 1$, group the leading terms as $x^5(252\pi x - 336\mathsf{A})$ and use $x^5 \geq x^3$ after the resulting coefficient is positive; indeed its rational lower bound is

$$252\frac{157}{50} - 336\frac{49}{40} = \frac{9492}{25} > 0.$$

Dropping the positive $24\mathsf{B}x$ term then gives

$$\begin{aligned} F_0'''(x) &> x^3\left(252\frac{157}{50} - 336\frac{49}{40} - 10\left(\frac{22}{7}\right)^2\right) - \frac{69}{14} \\ &\geq \frac{676141}{2450} > 0. \end{aligned} \tag{44}$$

Hence $F_0''(x) > 14$. With $y = x - 1 \geq 0$, Taylor's formula yields

$$F_0(x) > \frac{1}{4} - \frac{5}{2}y + 7y^2 = 7\left(y - \frac{5}{28}\right)^2 + \frac{3}{112} > 0. \tag{45}$$

A direct differentiation gives

$$\partial_K \underline{\mathcal{G}}(\rho, K_0) = D_0(x), \tag{46}$$

where

$$D_0(x) := \frac{\pi}{12}(3x^6 - 5 + 4x^{-6}) + \frac{1}{2}(x^3 - x^{-3}) - C_1x^2 + C_2x^{-2}. \tag{47}$$

At $x = 1$,

$$D_0(1) > \frac{1}{300}, \qquad D_0'(1) > \frac{54}{175}. \tag{48}$$

Moreover,

$$\begin{aligned} D_0''(x) &= \frac{\pi}{12}(90x^4 + 168x^{-8}) + 3x - 6x^{-5} - 2C_1 + 6C_2x^{-4} \\ &> \frac{15}{2}\frac{157}{50} - 3 - \frac{26}{25} = \frac{1951}{100} > 0. \end{aligned} \tag{49}$$

Here $x \geq 1$ gives $90x^4 \geq 90$ and $3x - 6x^{-5} \geq -3$; the omitted terms are nonnegative, and $-2C_1 > -26/25$. Thus the displayed lower bound follows term by term. Since $D_0''(x) > 0$,

$$D_0'(x) \geq D_0'(1) > 0 \qquad (x \geq 1).$$

Thus $D_0$ is increasing and $D_0(x) \geq D_0(1) > 0$.

Finally,

$$\partial_K^2 \underline{\mathcal{G}} = \frac{1 + 2\rho^2}{6} - \frac{4\alpha}{9}K^{-2/3} - \frac{2\beta}{9}K^{-4/3}. \tag{50}$$

For $K \geq K_0 \geq \pi$,

$$\alpha K^{-2/3} \leq \frac{\mathsf{A}}{\pi^2}, \qquad \beta K^{-4/3} \leq \frac{\mathsf{B}}{\pi^2}.$$

Consequently,

$$\partial_K^2 \underline{\mathcal{G}} > \frac{1}{6} - \frac{4(49/40) + 2(1/5)}{9(157/50)^2} = \frac{47447}{443682} > 0. \tag{51}$$

Since (51) is positive,

$$\partial_K \underline{\mathcal{G}}(\rho, K) \geq \partial_K \underline{\mathcal{G}}(\rho, K_0) = D_0(x) > 0 \qquad (K \geq K_0).$$

Hence

$$\underline{\mathcal{G}}(\rho, K) \geq \underline{\mathcal{G}}(\rho, K_0) = F_0(x) > 0 \qquad (K \geq K_0). \tag{52}$$

**Theorem 7** (Global low- and middle-ratio proxy theorem)**.** *For*

$$0 < \rho \leq \frac{39}{50}, \qquad K \geq \frac{\pi}{1 - \rho},$$

*one has*

$$P(\rho, K) < W(\rho, K). \tag{53}$$

*Consequently,* $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.

*Proof.* Equation (34), the angle bound, and (52) give

$$W - P > \underline{\mathcal{G}}(\rho, K) > 0.$$

The layer-cake and angular-block proofs retain every strict wall. □